CHAPTER 4: ELECTRIC POTENTIAL

Q.1 What is electric potential energy?

ELECTRIC POTENTIAL ENERGY

Consider a $+q_0$ test charge placed in electric field **E** created between two oppositely charged plates A and B. The electric field lines come out from positive plate and enter into negative plate.

The electric force acting on the test charge is given as:

$$\vec{F_e} = q_0 \vec{E_e}$$

The test charge $+q_0$ will move along electric lines of force and gains kinetic energy because positive plate B repels $+q_0$ charge while negative plate A attracts it.

The work done **W** by an external agent is needed to move test charge $+q_0$ with constant velocity against electric field between plate A and plate B respectively:

$$W = F_e \cdot ds$$

$$W = q_0 E \cdot ds$$

This negative work done is equal to difference in electric potential energy between two points:

$$U_b - U_a = -W$$

$$\Delta U = -W$$

Q.2 What is electric potential energy in terms of electric field?

Calculate electric potential energy of a system of charged particles:
The electric potential energy of a system of fixed point charges is equal to the amount of work done needed by an external agent to assemble the system. It is necessary that each charge is brought from infinite distance and the charges are at rest in their initial positions and in their final positions.

Consider a $+q_0$ test charge placed in an electric field created between two oppositely charged plates. The difference in potential energy ΔU when q_0 moves between point "a" and "b" under the action of electric force $F=q_0E$ is equal to negative work done and is given as:

$$\Delta U=-W_{ab}$$

This equation can be written as:

$$U_b-U_a=-\int_{ab}F_e\cdot ds$$

$$U_b-U_a=-\int_{ab}q_0E\cdot ds$$

When there is infinite separation between plates, the potential energy is zero at some reference point, so electric potential energy is written as:

$$U=-\int q_0E\cdot ds$$

[Image Placeholder 1]
Description: Likely shows a $+q_0$ test charge between two oppositely charged plates, illustrating the electric field and displacement.

Potential Energy of a System of Two Point Charges
Now consider two charges q_1 and q_2 separated by distance r as shown in fig.

[Image Placeholder 2]
Description: Shows two charges q_1 and q_2 separated by a distance r, likely illustrating the electric field lines or the displacement vector.

The change in potential energy when charge q_2 moves relative to $+q_1$ through displacement $ds=dr$ is:

$$U_b-U_a=-\int_{ab}q_2Edr$$

The electric field due to q1q_1 along the line connecting them is $E=r^2kq_1$. Therefore:

$$U_b - U_a = -\int_{ab} r^2 k q_1 q_2 dr$$

$$U_b - U_a = -k q_1 q_2 \int_{ab} r^{-2} dr$$

$$U_b - U_a = -k q_1 q_2 r^{-2+1}{}_{ab}$$

$$U_b - U_a = k q_1 q_2 [r_b 1 - r_a 1]$$

If we choose reference point 'a' at infinity ($U_\infty = 0$, $r_\infty = \infty$):

$$U_b - 0 = k q_1 q_2 [r_b 1 - 0]$$

$$U_b = r_b k q_1 q_2$$

The general formula for electric potential energy for two charges is:

$$U(r) = r k q_1 q_2$$

The electric potential energy is **negative** if the electric force is attractive (opposite signs of q_1 and q_2).

The electric potential energy is **positive** if the electric force is repulsive (same signs of q_1 and q_2).

Potential Energy of a System of Three Point Charges

Consider three charges q_1,q_2 and q_3 having separation r_{12},r_{13} and r_{23} as shown in fig.

[Image Placeholder 3]

Description: Likely shows a triangle formed by three point charges with labeled distances r_{12}, r_{13}, and r_{23}.

The electric potential energy of the system is:

$$U = U_1 + U_2 + U_3$$

$$U = r_{12} k q_1 q_2 + r_{13} k q_1 q_3 + r_{23} k q_2 q_3$$

Q.3 What is electric potential?

ELECTRIC POTENTIAL

Consider a $+q_0$ test charge placed in an electric field created between two oppositely charged plates A and B. The electric lines of force come out from the positive plate and enter into the negative plate.

The work done **W** by an external force is needed to move test charge $+q_0$ with constant velocity against the electric field between plate A and plate B respectively.

This negative work done is equal to difference in electric potential energy between two points:

$$\Delta U = U_b - U_a$$

$$\Delta U = -W_{ab}$$

The negative work done required to move a unit positive test charge with constant velocity from one point to another point is called **potential difference** between these points:

$$\Delta V = -q_0 W$$

$$V_b - V_a = -q_0 W$$

The electric potential Vb is zero when final position is shifted at infinity. Hence electric potential at any point in an electric field is:

$$\Delta V = q_0 W$$

UNIT OF ELECTRIC POTENTIAL

The SI unit of electric potential is the volt (V):

1volt=1joule/1coulomb

- The word voltage is often used for electric potential or potential difference.
- The change in potential energy is:

$$\Delta U = q0 \Delta V$$

- The unit of electric potential energy is the **joule (J)** when voltage is in volts and charge is in coulombs.
- A common unit for electric potential energy at the atomic scale is the electron volt (eV), where 1eV is the energy gained by an electron moving through a potential difference of 1 volt.

Q.4 Calculate electric potential due to point charge and n-point charges

ELECTRIC POTENTIAL DUE TO POINT CHARGE

Consider a $+q_0$ test charge placed in an electric field created between two oppositely charged plates A and B. The electric lines of force come out from the positive plate and enter into the negative plate.
The work done ***W*** on a test charge $+q_0$ by the electric force F_e when it moves through displacement ds from initial position i to final position f is:

$$W=\int_{if} F_e \cdot ds = q_0 \int_{if} E \cdot ds$$

$$q_0 W = \int_{if} E \cdot ds \cdots (1)$$

The **potential difference** between two points is defined as the negative work done per unit charge required to move it with constant velocity:

$$\Delta V = -q_0 W$$

$$q_0 W = -(V_f - V_i) \cdots (2)$$

Comparing equations (1) and (2):

$$V_f - V_i = -\int_{if} E \cdot ds$$

$$V_f - V_i = -\int_{if} E ds \cos\theta$$

For a radial path where E and ds are parallel ($\theta = 0_\circ$) and $ds = dr$:

$$V_f - V_i = -\int_{if} E dr$$

Setting Vf=0 at infinity and Vi=V at distance R:

$$0-V=-\int_{R}^{\infty}\frac{kq}{r^2}dr$$
$$V=\int_{R}^{\infty}kqr^{-2}dr$$
$$V=-kq[r^{-1}]_{R}^{\infty}$$
$$V=-kq(0-R^{-1})$$
$$V=\frac{kq}{R}$$

In general, for a point charge:

$$V=\frac{kq}{r}$$

- For a positive charge: $V=+\frac{kq}{r}$ $V=+\frac{kq}{r}$
- For a negative charge: $V=-\frac{kq}{r}$ $V=-\frac{kq}{r}$

This formula is used to calculate electric potential at any point in a given electric field.

Q. ELECTRIC POTENTIAL DUE TO COLLECTION OF N-POINT CHARGES

Consider charges $q_1, q_2, q_3 \cdots q_n$. Take a point P in the field of these charges having distances $r_1, r_2, r_3 \cdots r_n$.

[Image Placeholder 4]

Description: Likely shows multiple point charges and a point P with distances r_1, r_2, r_3... labeled.

The potential at a point due to a collection of n-point charges can be calculated by **superposition principle** which is stated as "*the potential at a point due to one of the charges is not affected by the presence of the other charges*."

$$V=V_1+V_2+V_3+\cdots+V_n$$

$$V=\sum_{i=1}^{n}V_i$$

$$V=\sum_{i=1}^{n}\frac{kq_i}{r_i}$$

Q.5 Explain how electric potential can be calculated from electric field

CALCULATION OF ELECTRIC POTENTIAL FROM ELECTRIC FIELD

Consider a $+q_0$ test charge placed in a non-uniform electric field E as shown in fig. Now take a non-linear path ab in this electric field.

[Image Placeholder 5]
Description: Shows a test charge $+q_0$ in a non-uniform field, following a curved path from point a to b.

The electric force acts on test charge $+q_0$. The force is $F_e = q_0 E$. The amount of work done dW_{ab} for small displacement ds is:

$$dW_{ab} = F_e \cdot ds = q_0 E \cdot ds$$

The net work done for path ab is:

$$W_{ab} = \int_{ab} q_0 E \cdot ds$$

Relation between Potential Difference and Potential Energy

The electric potential difference and electric potential energy are related as:

$$\Delta V = q_0 \Delta U \cdots (2)$$

Since $\Delta U = -W_{ab}$:

$$\Delta V = -q_0 W_{ab} \cdots (3)$$

$$W_{ab} = \int_{ab} q_0 E \cdot ds \cdots (1)$$

Substituting (1) into (3):

$$\Delta V = -\int_{ab} E \cdot ds$$

$$V_b - V_a = -\int_{ab} E \cdot ds$$

If point "a" is chosen as the reference point at infinity where $V_a = 0$:

$$V_b = -\int_{\infty b} E \cdot ds$$

This formula is used to calculate electric potential at any point in a given electric field.

Q. ELECTRIC POTENTIAL DUE TO CONTINUOUS CHARGE DISTRIBUTION

A collection of a large number of charges approximates a continuous charge distribution. There are three types:

1. Linear charge distribution
2. Surface charge distribution
3. Volume charge distribution

1. Electric Potential Due to Linear Charge Distribution

Consider a linear charge distribution. Take a small length element dz of this distribution having charge dq.

[Image Placeholder 6]

Description: Likely shows a line of charge with a small element dz and a point P at a distance r.

The linear charge density λ is charge per unit length:

$$\lambda = dzdq$$

The electric potential at P due to charge dq is:

$$dV = rkdq = rk\lambda dz$$

The net electric potential at P due to the entire distribution is:

$$V = \int rk\lambda dz$$

2. Electric Potential Due to Surface Charge Distribution

Consider a surface charge distribution. Take a small area element da of this distribution having charge dq.

[Image Placeholder 7]
Description: Likely shows a surface with a small area element da and a point P at a distance r.

The surface charge density σ is charge per unit area:

$$\sigma = \frac{dq}{da}$$

The electric potential at P due to charge dq is:

$$dV = \frac{k\sigma da}{r}$$

The net electric potential at P due to the whole surface is:

$$V = \int \frac{k\sigma da}{r}$$

3. Electric Potential Due to Volume Charge Distribution
Consider a volume charge distribution. Take a small volume element dV_0 of this distribution having charge dq.

[Image Placeholder 8]
Description: Likely shows a 3D volume with a small element dV_0 and a point P at a distance r.

The volume charge density ρ is charge per unit volume:

$$\rho = \frac{dq}{dV_0}$$

The electric potential at P due to charge dq is:

$$dV = \frac{k\rho dV_0}{r}$$

The net electric potential at P due to the whole volume is:

$$V = \int \frac{k\rho dV_0}{r}$$

Q. ELECTRIC POTENTIAL DUE TO A RING OF CHARGE

Consider a uniformly charged ring of radius R. Take a small length element ds of the ring having charge dq.

- The linear charge density is:

$$\lambda = dsdq, dq = \lambda ds, q = \lambda \int ds = \lambda(2\pi R)$$

[Image Placeholder 9]

Description: Shows a charged ring of radius R. A point P is on the z-axis at distance z from the ring's center. Distance r from ds to P is shown.

Now take a point P on the z-axis having distance z from the plane of the ring. Electric potential dV due to charge dq at point P having distance r is:

- Potential due to dq:

$$dV = rkdq = rk\lambda ds$$

From the figure,

$$r2 = R2 + z2.$$

Net potential at P due to the entire ring is:

$$V = \int rk\lambda ds = \int (R2 + z2)1/2 k\lambda ds$$

$$V = R2 + z2 k\lambda \int ds = R2 + z2 k\lambda(2\pi R)$$

$$V = R2 + z2 kq$$

This is the potential at a point on the axis of a ring at distance zz from its plane.

Q. BEHAVIOR OF A CHARGED DISK AT LARGE DISTANCE

For a charged disk, when the distance Z is much greater than the disk radius R ($Z >> R$), the potential approximates that of a point charge:

$$V = 4\pi\epsilon_0 Z\sigma(\pi R^2) = Zkq$$

Where $q = \int \sigma da = \sigma(\pi R^2)$. This shows a charged disk behaves as a point charge in the limiting case when $Z >> R$.

Q.11 What are equipotential surfaces?

Draw the lines of force & cross sections of equipotential surfaces for:

(i) a uniform field (ii) a positive point charge

Q. EQUIPOTENTIAL SURFACES

A family of surfaces connecting points having the same electric potential is called **equipotential surfaces**.

1. For a Uniform Electric Field

Consider a uniform electric field E. It is indicated by horizontal electric lines of force as shown in fig. The perpendicular dashed lines are cross-sections of equipotential surfaces.

[Image Placeholder 10]

Description: Shows a uniform electric field (horizontal arrows) with vertical dashed lines representing equipotential surfaces. Points A, B, C, D, E are labeled.

Take two points A and B separated by distance L. The potential difference is:

$$\Delta V = -\int_{AB} E \cdot ds = -\int_{AB} E ds \cos 0^\circ = -EL$$

The magnitude of work done by the electric field on a unit positive test charge q_0 moved from A to B is $q_0 EL$.

Consider points B and C which lie on the same equipotential surface. The potential difference is:

$$\Delta V = -\int_{BC} E \cdot ds = -\int_{BC} E ds \cos 90^\circ = 0$$

Thus:

$$V_C - V_B = 0 \Longrightarrow V_C = V_B$$

Since $\Delta V = -_{q_0} W_{BC}$, putting $\Delta V = 0$ gives $W_{BC} = 0$. Hence no work is done by the electric field when q0q_0 is moved from B to C. All points on the line containing B, C, D, etc., have the same potential. The surfaces passing through these points are equipotential surfaces.

2. For a Positive Point Charge

Now consider a $+q$ charge placed at the center of a sphere. The electric field lines are along the radial direction as shown in fig.

[Image Placeholder 11]

Description: Shows a point charge at the center of concentric circles (spheres in 3D) representing equipotential surfaces. Radial field lines are shown.

The potential difference between two points A and B is:

$$V_A - V_B = kq(r_A 1 - r_B 1)$$

When points A and B lie on the surface of the same sphere ($r_A = r_B$):

$$V_A - V_B = kq(r_A 1 - r_A 1) = 0 \Longrightarrow V_A = V_B$$

All points at a given radius have the same potential. Therefore, the equipotential surfaces of a point charge form a family of **concentric spheres.** For a dipole, the equipotential surfaces are more complicated.

Features of Equipotential Surfaces

A figure shows a family of equipotential surfaces.

[Image Placeholder 12]

Description: Likely shows a set of nested, non-uniform equipotential surfaces (like those

around a dipole or irregular charge distribution). Points 1, 2, 3, 4 are marked on different surfaces.

It has the following **features**:

1. The work done by the electric field when a charge moves from point 1 to point 2 is zero because both paths start and end on the same equipotential surface.
2. Moving from point 3 to point 4, work done is non-zero because the path starts on one surface and ends on another.
3. Equipotential surfaces are always **perpendicular** (at right angles) to the electric field lines.

Q.12 Explain how electric field is calculated from given electric potential or prove that negative of potential gradient is equal to electric field (E = -grad V)

ELECTRIC FIELD AND POTENTIAL GRADIENT ($E=-\nabla V$)

The **potential gradient** at a point is the maximum rate of increase of electric potential with respect to distance. The electric field is related to the potential by:

$$E=-\nabla V=-\text{grad}V$$

Where ∇ (or grad) is the gradient operator.

Derivation

Consider a cross-section of a family of equipotential surfaces differing in potential by dV as shown in fig.

[Image Placeholder 13]

Description: Shows several closely spaced equipotential lines (V, V+dV, V-dV). A point P on surface V has a test charge q_0. The electric field E and a small displacement ds at an angle θ are shown.

The electric field E at point P is perpendicular to the equipotential surface passing through P.

The work done dW by the electric field on test charge $+q_0$ when it is moved from the surface at potential V to the surface at potential $V+dV$ through displacement ds is equal to the negative change in potential energy:

$$dW=-q_0dV\cdots(1)$$

Expressing work in terms of force and displacement:

$$dW=F\cdot ds=q_0E\cdot ds=q_0Eds\cos\theta\cdots(2)$$

Equating (1) and (2):

$$q_0Eds\cos\theta=-q_0dV$$

$$E\cos\theta=-dsdV$$

This states that the component of E in any direction is equal to the negative rate of change of potential in that direction. The negative sign means E points in the direction of decreasing V.

The maximum value occurs when $\cos\theta=1$ (i.e., when ds is in the direction of the maximum rate of increase of V):

$$E=-(dsdV)_{\max}$$

The maximum value of $dsdV$ is defined as the **potential gradient**, $\text{grad}V$:

$$E=-\text{grad}V$$

$$E=-\nabla V$$

Component Form in Cartesian Coordinates

$$E=E_xi\hat{}+E_yj\hat{}+E_zk\hat{}=-(\partial x\partial Vi\hat{}+\partial y\partial Vj\hat{}+\partial z\partial Vk\hat{})$$

Therefore, by comparing coefficients:

$$E_x = -\partial x \partial V, E_y = -\partial y \partial V, E_z = -\partial z \partial V$$

Example: Electric Field from Quadrupole Potential

The electric potential due to a quadrupole varies as $r_3 1$. The potential at a distant point is:

$$V = r_3 kQ = kQr_{-3}$$

The electric field is found from the potential gradient:

$$E = -drdV = -drd(kQr_{-3}) = 3kQr_{-4}$$

$$E = r_4 3kQ$$

CHAPTER 8: MAGNETISM FUNDAMENTALS

Q. What is magnetism? State its various properties and uses.

MAGNETISM

Magnetism is a physical phenomenon produced by the motion of electric charge, resulting in attractive and repulsive forces between objects. It was known to ancient civilizations (Greeks, Romans, and Chinese) through the mineral **lodestone** (magnetite), a naturally occurring iron oxide that attracts iron objects.

A **magnet** is a body that attracts certain materials like iron and, when freely suspended, aligns itself approximately north-south. The ability to exert these forces is called **magnetization**. The branch of physics that studies these forces and associated fields is called **magnetism**.

TYPES OF MAGNETS

Magnets are broadly classified into two types:

1. **Permanent Magnets**
2. **Electromagnets**

1. PERMANENT MAGNETS

These are made from materials (like hard steel, alnico, or ferrite) that retain their magnetism once magnetized. They can occur naturally (lodestone) or be manufactured.

PROPERTIES OF PERMANENT MAGNETS:

1. **Poles:** When dipped into iron filings, a bar magnet attracts most filings at its ends, known as **poles**. The middle (neutral region) shows little attraction.
2. **Dual Nature:** Every magnet has two inseparable poles—a **North (N) Pole** and a **South (S) Pole**.

- **Unlike poles attract** (N and S).
- **Like poles repel** (N-N or S-S).

3. **Indivisibility of Poles:** If a magnet is broken, each piece becomes a new magnet with its own N and S poles. Magnetic poles cannot be isolated as single entities.
4. **Magnetization & Demagnetization:** An iron or steel bar can be magnetized by stroking it with a magnet. Magnetization can be weakened or removed (demagnetization) by heating, hammering, or dropping the magnet.
5. **Magnetic Field:** The region surrounding a magnet where its magnetic force can be detected is called the **magnetic field (B)**. It is a vector quantity represented by **magnetic field lines**, which emerge from the N pole and enter the S pole.

2. ELECTROMAGNETS

An electromagnet is a temporary magnet created by passing an electric current through a coil of wire (solenoid), often wound around a soft iron core. The magnetic field exists only while current flows.

ELECTROMAGNETISM is the study of the relationship between electricity and magnetism.

Properties of an Electromagnetic Field:

- **Magnitude:** The strength of the magnetic field depends on the amount of current and the number of turns in the coil, as described by **Ampere's Law**.
- **Direction:** The direction of the magnetic field around a current-carrying conductor is given by the **Right-Hand Rule**.

RIGHT-HAND RULE:
For a straight conductor, grasp it with your right hand so that your thumb points in the **direction of conventional current flow** (positive to negative). Your curled fingers will then point in the **direction of the concentric magnetic field lines** around the conductor.

USES OF MAGNETS

1. **Everyday Applications:** From refrigerator door seals and magnetic latches to speakers, headphones, and credit/debit card stripes.
2. **Data Storage:** Used in hard disk drives, magnetic tapes, and floppy disks for recording and storing digital information.
3. **Medical Imaging: MRI (Magnetic Resonance Imaging)** uses the magnetic properties of atomic nuclei to create detailed images of internal organs and tissues.
4. **Scientific Research:** Spacecraft measure planetary magnetic fields to study internal structures. Particle accelerators like **cyclotrons** use powerful electromagnets to control charged particle beams.

5. **Industrial & Lifting:** Powerful electromagnets are used in cranes to lift and move heavy scrap iron, steel, or vehicles.
6. **Medical Procedures:** Electromagnets can be used to safely remove small ferromagnetic objects (e.g., iron filings from eyes) from the body.
7. **Electromechanical Devices:** Essential components in **electric bells, motors, generators, transformers, relays, galvanometers, and moving-coil meters**.
8. **Communication & Electronics:** Found in telephones, radios, television sets, and microphones.

MAGNETIC FORCE

Q 2. Define a formula to: magnetic force acting on current carrying conductor placed in aluminum field.

MAGNETIC FORCE ON CURRENT CARRYING CONDUCTOR PLACED IN A MAGNETIC FIELD

Consider a conductor of length L is placed over a conducting rail in applied magnetic field B perpendicularly. The magnetic field is generated around the conductor in the form of circles when current I flows through conductor. The generated magnetic field and applied magnetic field interacts together and give rise to magnetic force.

EXPERIMENTAL OBSERVATIONS:

It is experimentally observed that motion of conductor on rail becomes fast when amount of current through circuit or applied magnetic field increases. It is also observed that motion of conductor increases when length of conductor increases. Similarly, motion of conductor is also affected with its orientation to magnetic filed.

Magnetic Force Formula:

This magnetic force F_m acting on conductor is directly proportional to current I and length L. The magnetic force is also proportional to applied magnetic field B and orientation (Sinθ) of conductor with magnetic field.

$$F_m \propto L$$

$$F_m \propto B$$

$$F_m \propto Sin\theta$$

On combining al relations

$$F_m \propto ILBSin\theta$$

$$F_m = kILBSin\theta$$

Where k is constant of proportionality. Its value in SI system is unity.

$$F_m = ILBSin\theta$$

In **vector form**, the magnetic force is

$$F_m = I(L \times B)$$

$$F_m = ILBSin\theta n\hat{}$$

Where L is called **vector length** of conductor whose magnitude is the length of the conductor and direction is the direction of current.

FORCE BETWEEN PARALLEL CONDUCTORS

This is magnitude of magnetic force which is acting on conductor P. The direction of this magnetic force $F_{12} = I_1(L \times B_2)$ is away from conductor Q according to right hand rule. Comparing eq(1) and eq(2)

$$F_{12} = F_{21}$$

It shows that F_{12} and F_{21} are equal in magnitude but acting away from each other and their point of application is not same. Therefore, it is proved that antiparallel currents repel each other.

FORCE IN NON_UNIFORM FEILD

When wire is not straight or magnetic field is non-uniform, then to calculate magnetic force, divide the wire into small segments of length dS for which length becomes uniform approximately.

The magnetic force on each segment is

$$dF_m = I(dS \times B)$$

The net magnetic force on conductor having length L is

$$F_m = \int_{0L} I(dS \times B)$$

DIRECTION OF MAGNETIC FORCE

The direction of magnetic force is given by right hand rule stated as:
Place close-fisted right hand at a location where tails of length vector L and magnetic field vector are joined together in such a way that fingertips are along the direction of rotation through smaller angle, then stretched thumb indicates the direction of magnetic force.

MAGNETIC FORCE ON CURRENT CARRYING CONDUCTOR

The magnetic force F_L on current carrying conductor having length vector L placed in applied magnetic field B is

$$F_L = I(L \times B)$$

$$F_L = nAqV(L \times B)$$

Where $L\hat{} = V\hat{}$ because direction of length vector and velocity vector is same.

$$F_L = nAqV(LV\hat{} \times B)$$

$$F_L = nAqL(VV\hat{} \times B)$$

$$F_L = nAqL(V \times B)$$

MAGNETIC FORCE ON SINGLE CHARGE

The magnetic force F_m on a single charge q moving with velocity V in applied magnetic field B:

$$F_m = nALF_L$$

$$F_m = nAqL(V \times B)$$

$$F_m = q(V \times B)$$

This is magnetic force on single charge which enters in applied magnetic field can be written as

$$F_m = qVB\sin\theta n\hat{}$$

MAGNITUDE OF MAGNETIC FORCE

The magnitude of magnetic force is

$$F_m = qVB\sin\theta$$

The magnetic force is maximum when angle between V and B is 90°.

$$F_{max} = qVB\sin 90^\circ$$

$$F_{max} = qVB$$

The magnetic force is minimum when angle between V and B is 0°

$$F_{min} = qVB\sin 0^\circ$$

$$F_{min} = 0$$

Similarly, magnetic force is minimum when angle between V and B is 180°

$$F_{min} = qVB\sin 180^\circ$$

$$F_{min}=0$$

DIRECTION OF MAGNETIC FORCE

The direction of magnetic force F_m is given by right hand rule stated as "join the tails of velocity vector V of +q charge and magnetic field vector B together. This defines a plane. Place close-fisted right hand at a location where tails of two vectors be joined in such a way that finger tips are along rotation through smaller angle.

The erect thumb indicates the direction of magnetic force.

$$F_m=q(V\times B)$$

The direction of magnetic force is reversed when charge is negative.

SECOND DEFINITION OF RIGHT HAND RULE

The right hand rule can also be stated as

- Point the thumb of your right hand in the direction of velocity V of +q charge
- Fingers in the direction of magnetic field B.
- Palm direction is the direction of magnetic force

$$F_m=q(V\times B)$$

MAGNETIC FORCE ON ELECTRON

The magnetic force acting on an electron entering with velocity V in applied magnetic field B is

$$F_m=-e(V\times B)$$

MAGNETIC FORCE ON PROTON

The magnetic force acting on proton entering with velocity v in applied magnetic field B is

$$F_m=+e(V\times B)$$

When an electron and proton are projected into magnetic field in the same direction both are deflected in opposite direction because magnetic force acts on them in opposite direction.

Q. Discuss motion of charged particle in an electric field and magnetic field hence define Lorentz force?

MOTION OF CHARGED PARTICLE IN AN ELECTRIC FIELD

Consider a +q charge.is placed in a uniform electric field. The electric force acting on the charge is given as

$$F_e = qE \cdots\cdots (1)$$

The charge +q will move parallel to electric field E if it is free to move. The acceleration a developed is given as

$$F_e = ma \cdots\cdots (2)$$

Comparing eq(1) and eq(2)

$$\mathrm{ma} = \mathrm{qE}$$

$$ma = qE$$

$$a = mqE$$

The position of charge at any time can be calculated by using three equations of motion. The electric force always do work on charge when placed inside electric field.

MOTION OF CHARGED PARTICLE IN MAGNETIC FIELD

The magnetic force F_m acts on the charge when a charge +q enters into magnetic field B with velocity V is given as

$$F_m = +q(V \times B)$$

- The maximum magnetic force acts on +q charge when it enters into magnetic field perpendicularly.

- The charge moves in **circular path** under the action of this magnetic force.
- The magnetic force which is always perpendicular to the plane defined by V and B is only deflecting force.
- It does not perform any work.

$$W = F_m \cdot S$$

$$W = F_m \cdot Vt$$

$$W = FVt\cos 90^\circ$$

$$W = 0$$

LORENTZ FORCE

The total force acting on a charge is called **Lorentz force** when it enters in a region where electric field and magnetic field are applied at the same time. The value of Lorentz force is given as

$$F = F_c + F_m$$

$$F = qE + q(V \times B)$$

Q. Discuss application of Lorentz force in velocity selector.

LORENTZ FORCE AND VELOCITY SELECTOR

A **velocity selector** is a device that uses crossed electric and magnetic fields to allow only particles of a certain velocity to pass through undeflected.

- Suppose a positively charged particle enters a region with both electric field E and magnetic field B applied perpendicular to each other.
- Electric force:

$$F_e = qE$$

- Magnetic force:

$$F_m = qVB$$

For the particle to pass undeflected:

$$F_e = F_m$$
$$qE = qVB$$
$$E = VB$$

Thus, only particles with velocity $E=VB$ will pass straight through.

Application:

- Used in particle accelerators.
- Used in mass spectrometers to select particles of specific velocity.
- The particle accelerator is an application of Lorentz force.

CHAPTER 9: AMPERE'S LAW

Q6.3 Size and explain Ampere law, write its integral and differential form

AMPERE'S LAW

The Ampere law states that a magnetic field is generated around a conductor when current flows through it. The magnitude of generated magnetic field at a point is directly proportional to amount of current and inversely proportional to distance between point and straight current carrying conductor.

DIRECTION OF MAGNETIC FIELD

The direction of magnetic field is given by **right hand rule** stated as

Grasp the current carrying wire in your right hand in such a way that erect thumb is along the direction of current. The curled fingertips around the wire indicates the direction of magnetic field.

EXPLANATION

Consider a straight conductor through which current I flows. The magnetic field B is generated around it in the form of circles.

The magnitude of magnetic field B depends upon current I and distance r between conductor and point P where we want to evaluate magnetic field.

$$B \propto I \qquad B \propto r1$$

Combining both experimental facts

$$B \propto r1$$

$$B = 2\pi\mu_0(r1)$$

Where $2\pi\mu_0$ is constant of proportionality and μ_0 is called permeability of free space. Its value is $4\pi \times 10^{-7}$Wb/A-m. It is the degree of magnetization that a material obtains in response to an applied magnetic field.

Where $2\pi r$ is the length of path called circumference of circle and denoted by t. The value of B is constant around the path

$$B\ell = \mu_0 I$$

$$B\ell\cos 0^\circ = \mu_0 I$$

$$B \cdot \ell = \mu_0 I$$

Consider an amperian loop to find magnetic field set up by current in a long straight wire such that wire is perpendicular to plane of loop and current flows out of wire as shown in fig.

Take a small length element dr of the amperian loop. The total line integral for amperian loop is

$$\oint B \cdot dr = \mu_0 I$$

$$\oint B dr \cos\theta = \mu_0 I$$

AMPERE'S CIRCUITAL LAW

For any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.

INTEGRAL FORM OF AMPERE'S LAW

$$\oint B \cdot dr = \mu_0 I$$

The value of current in terms of current density is

$$I = \int J \cdot da$$

The above eq becomes

$$\oint B \cdot dr = \mu_0 \int J \cdot da$$

This is called integral form of Ampere's law

DIFFERENTIAL FORM

The integral form of Ampere's law is

$$\oint B \cdot dr = \mu_0 \int J$$

$$\int (\nabla \times B - \mu_0 J) \cdot da = 0$$

Since da cannot be zero so

$$\nabla \times B - \mu_0 J = 0$$

$$\nabla \times B = \mu_0 J$$

This is called differential form of Ampere's law.

Q6.8 Apply Ampere's law to calculate magnetic field within and outside a current carrying solenoid?

SOLENOID

A long tightly wound cylindrical conducting wire is called solenoid. The magnetic field is generated when current flows through windings of solenoid. It provides uniform magnetic field and used to store magnetic energy.

- The magnetic field inside solenoid is uniform and strong called strong magnetic field.
- The magnetic field outside the solenoid is weak.
- The iron core solenoid has much greater magnetic field as compared to the air core solenoid.

MAGNETIC FIELD INSIDE SOLENOID

Consider a solenoid having N turns and length ℓ. The current I_0 flows through each turn of solenoid. The turns per unit length of solenoid are $n=\ell N$. The strong magnetic field B is generated inside solenoid which is directed along axis of solenoid. The magnetic field outside solenoid is so weak that it is considered zero.

Take amperian loop in the shape of rectangle to calculate magnetic field inside solenoid by applying Ampere's law

$$\oint B\cdot dr=\mu_0 I$$

$$\int B\cdot dr+\int B\cdot dr+\int B\cdot dr+\int B\cdot dr=\mu_0 I$$

$$\int Bdr\cos 0^\circ+\int Bdr\cos 90^\circ+\int B_{out}dr\cos 180^\circ+\int Bdr\cos 90^\circ=\mu_0 I$$

$$\int Bdr(1)+\int Bdr(0)+\int(0)dr(-1)+\int Bdr(0)=\mu_0 I$$

$$\int Bdr=\mu_0 I$$

The length of loop from a to b is $h\sin\theta$, $\mathrm{dr}=h$

$$Bh=\mu_0 I$$

Unit length of solenoid has turns = n

h length of loop intercept turns = nh

TOTAL CURRENT IN TURNS OF SOLENOID

One turn of solenoid has current = I_0

nh turns of solenoid have current = nhI_0

$$B=\mu_0 nI_0$$

This is the magnitude of magnetic field inside solenoid.

DIRECTION OF MAGNETIC FIELD

The direction of magnetic field is given by right hand rule stated as "hold the solenoid in your right hand in such a way that fingertips are along direction of current then erect thumb indicates the direction of magnetic field which is along axis of solenoid.

The solenoid acts as **bar magnet**. The thumb side is north pole and other side is south pole. The magnetic field lines come out from north pole and enter into south pole.

MAGNETIC FIELD OUTSIDE SOLENOID

The solenoid acts as bar magnet when current flows through the windings of solenoid. The magnetic field lines come out from north pole and enter into south pole. These magnetic field lines are concentrated inside the solenoid while spread in space outside the solenoid. That is why magnetic field outside solenoid becomes weak.

Consider a solenoid having n turns per unit length. The current I_0 flows through each turn of solenoid. To calculate magnetic field outside solenoid, imagine an amperian loop having radius r which encloses one turn. The magnetic field lines will be tangential.

The Ampere's law is

$$\oint B \cdot dr = \mu_0 I$$

$$\oint B_t dr \cos 0^\circ = \mu_0 I_0$$

$$B_t \oint dr = \mu_0 I_0$$

$$B_t (2\pi r) = \mu_0 I_0$$

$$B_t = 2\pi r \mu_0 I_0 \cdots\cdots (1)$$

This is magnetic field outside the solenoid. The magnetic field inside the solenoid is

$$B = \mu_0 n I_0 \cdots\cdots (2)$$

Divide eq(1) and eq(2)

$$BB_t=(2\pi r\mu_0 I_0)(\mu_0 n I_0 1)$$

$$B_t=2\pi rnB$$

It shows B_t is smaller than B because $2\pi m$ is very large factor. That is why magnetic **field outside solenoid is weaker than inside.**

Q. Apply Ampere's law calculate magnetic field at the interior points of current carrying toroid.

TOROID

The solenoid bent into circular shape is called **toroid**. A strong magnetic field is generated when current flows through windings of a solenoid. These magnetic field lines are along the axis of the solenoid. These magnetic field lines become concentric circles in toroid.

The toroid is a very useful device used in tape heads and tokamaks. It is central feature of the tokamak which is basis of fusion power reactor.

MAGNETIC FIELD DUE TO TOROID

The hollow space of toroid has no magnetic field. There is no magnetic field around the windings of toroid because magnetic field generated between two adjacent turns is cancelled out. Therefore, magnetic field outside toroid is zero.

All the magnetic field is present inside the tubular path of toroid. The magnetic field is not constant over the entire cross section of the toroid.

Consider a toroid having N turns and each turn carries current I_0. The total current is given as

$$I=NI_0$$

To calculate magnetic field B at a point P inside toroid, imagine an amperian loop of radius r such that point P lies on it. The magnetic field B and small length element dr of amperian loop are along tangent direction so makes zero angle.

Apply Ampere's law

$$\oint B \cdot dr = \mu_0 I$$

$$\oint B dr \cos 0^\circ = \mu_0 N I_0$$

$$B \oint dr = \mu_0 N I_0$$

$$B(2\pi r) = \mu_0 N I_0$$

$$B = 2\pi r \mu_0 N I_0$$

This is magnitude of magnetic field inside toroid. It depends upon turns of toroid and radius r.

DIRECTION OF MAGNETIC FIELD

The direction of magnetic field within toroid is given by right hand rule stated as "cut the fingers of your right hand in the direction of the current, then extended thumb indicates the direction of the magnetic field". These filed lines are circular.

CHAPTER 10: FARADAY'S LAW & ELECTROMAGNETIC INDUCTION

Q. **Q1. What is motional emf? Derive its formula. Calculate electrical power exerted on a loop of wire.**

MOTIONAL EMF

The emf induced in a conductor when it is moved across a magnetic field is called **motional emf**.

DERIVATION

Consider a conductor of length D placed on a conducting rail. The ends c and d of the rail are connected with a galvanometer. A magnetic field is applied perpendicularly into the plane of the rail.

- When the rod is moved with constant velocity, the galvanometer shows deflection → emf is induced.
- A charge +q inside the rod moves with velocity $\vec{V}$ when rod is moved across the magnetic field and experiences a magnetic force:

$$\vec{F}m=q(\vec{V}\times\vec{B})$$

For example:

$$\vec{F}m=qVBj\hat{}$$

This force pushes +*q* charges towards one end of the rod and -*q* charges towards the opposite end.

INDUCED ELECTRIC FIELD

The induced electric field in the rod is:

$$\vec{E}=\vec{F}mq=VBj\hat{}$$

Magnitude:

$$E=VB$$

EMF

Electromotive force (emf) is related to electric field:

$$\varepsilon=\int\vec{E}\cdot d\vec{r}$$

$$\varepsilon=\int E\,dr\cos\theta$$

$$\varepsilon=\int 0DVB\,dr$$

$$\varepsilon=VBD$$

Thus, motional emf:

$$\varepsilon=VBD$$

The motional emf $\varepsilon=VBD$ is induced when rod is moved across the magnetic field towards right with constant velocity V.

INDUCED CURRENT

The amount of current induced in the loop having resistance R due to motional emf is given as

$$I=R\varepsilon$$

$$I=RVBD$$

Magnetic Force on Current-Carrying Rod

Now consider a rod of length D carrying current I in applied magnetic field $\vec{B}$.

- Magnetic force:

$$F_L=I(\vec{L}\times\vec{B})$$

For length vector $\vec{L}=Dj\hat{}$:

$$F_L=IDB(-i\hat{})$$

Magnitude:

$$F_L=IDB$$

Substituting induced current:

$$F_L=VBDR\cdot DB=RV^2B^2D^2$$

MECHANICAL POWER

The rate of doing working is called **mechanical power** and given as

$$P_m=F_LV$$

$$P_m=(RVB^2D^2)V$$

$$P_m=RV^2B^2D^2\cdots\cdots(1)$$

ELECTRICAL POWER

The rate of energy dissipated in the loop having resistance R due to induced current "I" is called **electrical power** P_e

$$P_e = i^2 R$$

$$P_e = (RVBD)^2 R$$

$$P_e = RV^2 B^2 D^2 \cdots\cdots (2)$$

CONCLUSION

$$Pm = Pe$$

This means **mechanical power is completely converted into electrical power**. The principle of motional emf is the **basis of AC generators**, which convert mechanical energy into electrical energy.

FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

The current induced in loop must flow either in clockwise direction or anticlockwise direction. The Lenz's law decides the direction of current.

The direction of induced current must be anticlockwise when north pole of bar magnet is moved towards the coil. In this situation face of coil facing bar magnet becomes north pole which repels north pole of bar magnet so motion of bar magnet is opposed.

Similarly direction of induced current becomes clockwise when north pole of bar magnet is pulled out from coil. In this situation face of coil facing bar magnet becomes south pole which is attracted by north pole of bar magnet so motion of bar magnet is again opposed.

The "push" of the magnet is the "cause" that produces the induced current because magnetic field generated due to induced current opposes the motion of bar magnet.

Similarly "pull" of the magnet is the "cause" that produces the induced current because magnetic field generated due to induced current opposes the motion of bar magnet.

Q. LENZ'S LAW AND LAW OF CONSERVATION OF ENERGY

The Lenz's law is in accordance to law of conservation of energy because mechanical energy is converted into electrical energy when a conductor is moved across the magnetic field but total energy is constant.

Consider a conductor having length D is placed on a conducting rail whose ends are connected with galvanometer. The magnetic field $\mathbf{B}=-\mathbf{Bk}$ is directed into plane of paper and denoted by x.

EMF

Electromotive force (emf) is related to electric field:

$$\varepsilon=\int \vec{E}\cdot d\vec{r}$$

$$\varepsilon=\int E\, dr\cos\theta$$

$$\varepsilon=\int 0DVB\, dr$$

Thus, motional emf:

$$\varepsilon=VBD$$

The motional emf $\varepsilon=VBD$ is induced when rod is moved across the magnetic field towards right with constant velocity V.

INDUCED CURRENT

The amount of current induced in the loop having resistance R due to motional emf is given as

$$I=R\varepsilon$$

$$I=RVBD$$

The direction of induced current in the loop is such as it opposes the cause producing it. The direction of induced current in the loop is anticlockwise according to

Lenz's law. Now rod of length D carrying current I is present in applied magnetic field B = - B k.

Magnetic force F_L acting on the rod having length vector L=Dj$L=D\mathbf{j}$ is given as

$$F=I(L\times B)F=-IDB(j\times k)F=IDB(-i)$$

Magnetic force acting on current carrying rod is in opposite direction to applied external force.

Magnitude of this magnetic force is given as

$$F_L=IDB$$

$$F_L=RVBD(DB)$$

$$F_L=RVB_2D_2$$

MECHANICAL POWER

The rate of doing working is called mechanical power and given as

$$P_m=F_LV$$

$$P_m=(RVB_2D_2)V$$

$$P_m=RV_2B_2D_2\cdots\cdots(1)$$

ELECTRICAL POWER

The rate of energy dissipated in the loop having resistance R due to induced current "I" is called electrical power P_e

$$P_e=i_2R$$

$$P_e=(RVBD)_2R$$

$$P_e=RV_2B_2D_2\cdots\cdots(2)$$

CONCLUSION

The above eq(1) and eq(2) gives $P_m = P_e$. It means mechanical power is completely converted into electrical power.

Hence Lenz's law is in accordance with law of conservation of energy.

Q. DERIVATION OF FARADAY'S LAW

The emf induced in rod when it is pulled towards right with constant velocity. V is given as

$$\varepsilon = -VBL$$

Velocity V of rod when it travels distance dx in time dt is

$$V = dtdx$$

Put this value in above eq

$$\varepsilon = -(dtdx)BL$$

$$\varepsilon = -dtBda[Ldx = da]$$

$$\varepsilon = -dtd\phi_m$$

Induced emf when loop has N turns

$$\varepsilon = -Ndtd\phi_m \cdots (1)$$

The negative sign means direction of induced emf is such as it opposes the cause producing it.

Magnetic flux $d\phi_m$ is given as

$$d\phi_m = B \cdot da$$

Net magnetic flux is

$$\phi_m = \int B \cdot da$$

Put this value in eq (1)

$$\varepsilon = -N\frac{d}{dt}\int B \cdot da \cdots (2)$$

INTEGRAL FORM OF FARADAY'S LAW

The mathematical form of Faraday's law is

$$\varepsilon = -N\frac{d}{dt}\int B \cdot da \cdots (3)$$

Emf and electric field are related as

$$\varepsilon = \int E \cdot dr \cdots (4)$$

SUMMARY OF ORGANIZATION:

1. **Section 1: Magnetism Fundamentals** (Chapters 8)
2. **Section 2: Magnetic Force**
3. **Section 3: Electric Potential** (Chapter 4) - **NEWLY ADDED**
4. **Section 4: Ampere's Law** (Chapter 9)
5. **Section 5: Faraday's Law & Electromagnetic Induction** (Chapter 101)
6. **Section 6: Force Between Parallel Conductors**